

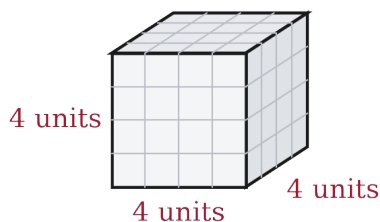
Cube ValuesWorksheet 7A.02.03 **ANSWER KEY**

1. Complete the table. Each volume is the side length multiplied by itself three times.

Side s	1	2	3	4	5
Volume s^3	1	8	27	64	125
Side $-s$	-1	-2	-3	-4	-5
Volume $(-s)^3$	-1	-8	-27	-64	-125

The bottom two rows are the point of the table: a negative side length is not a real box, but $(-s)^3$ is a real number and it is negative. Squares never do this. Come back to it in question 13.

2. Find the volume of the cube shown.



$$V = 4^3 = 4 \times 4 \times 4 = 64 \text{ cubic units}$$

A student who answers 12 has multiplied by 3 instead of cubing. A student who answers 16 has stopped after squaring.

3. A cube-shaped box has a volume of 125 cubic inches. How long is one edge?

$$5 \times 5 \times 5 = 125, \text{ so each edge is 5 inches.}$$

125 is the largest perfect cube the Grade 8 FAST will ask about, and -125 is the smallest.

For questions 4 through 7, find the value.

4. $\sqrt[3]{27}$

3 $3 \times 3 \times 3 = 27$

5. $\sqrt[3]{-64}$

-4 $(-4) \times (-4) \times (-4) = -64$

Unlike a square root, a cube root of a negative number does exist. Three negative factors give a negative product.

6. $\sqrt[3]{125}$

5

7. $\sqrt[3]{-1}$
 -1

For questions 8 through 10, find every real solution of the equation.

8. $x^3 = 8$

$x = 2$ one solution, not two

$(-2)^3 = -8$, not 8, so -2 is not a solution. This is the single most-missed idea on this benchmark.

9. $y^3 = -125$

$y = -5$

10. $n^3 = 0$

$n = 0$

11. Select ALL of the values below that make $x^3 = 64$ true.

☒ 4

☐ -4

☒ $\sqrt[3]{64}$

☐ $-\sqrt[3]{64}$

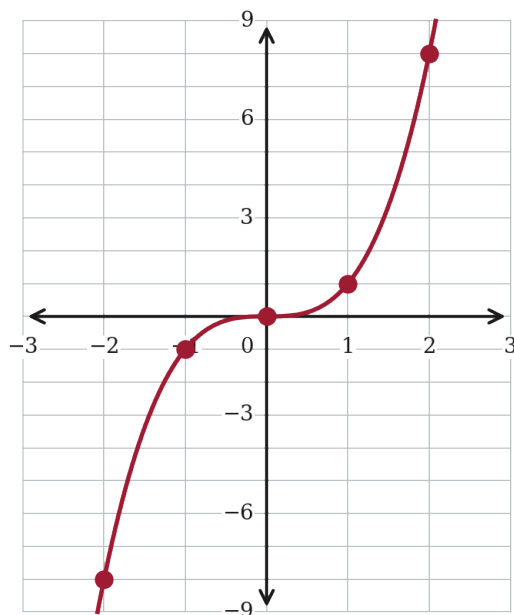
☐ 8

☐ $21\frac{1}{3}$

Two of the six. This exact question appeared on the 2025 released items and 26% of students got it right — one of the six lowest scores on the whole exam. The trap is -4 and $-\sqrt[3]{64}$, both of which give -64 . 8 is the square root of 64; $21\frac{1}{3}$ is $64 \div 3$.

12. Complete the table of values for $y = x^3$. Then plot the five points on the grid.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2	$y = (-2)^3 = -8$	$(-2, -8)$
-1	$y = (-1)^3 = -1$	$(-1, -1)$
0	$y = (0)^3 = 0$	$(0, 0)$
1	$y = (1)^3 = 1$	$(1, 1)$
2	$y = (2)^3 = 8$	$(2, 8)$



Compare with the graph of $y = x^2$ from Lesson 1. That one stayed on or above the x -axis. This one crosses it, because cubing a negative gives a negative.

13. The equation $x^2 = 64$ has two solutions, but the equation $x^3 = 64$ has only one. Explain why.

$x^2 = 64$ has two solutions because $8 \times 8 = 64$ and $(-8) \times (-8) = 64$. Two negative factors multiply to a positive.

$x^3 = 64$ has one solution because $4 \times 4 \times 4 = 64$, but $(-4) \times (-4) \times (-4) = -64$. Three negative factors multiply to a negative.

So a positive number has two square roots but only one cube root.

The point needs the count of negative factors, not just the answers. A student who says “because cubes are different” has not explained anything.

STRETCH. The two questions below use cubes larger than the Grade 8 FAST will ask for. The method is identical — only the arithmetic is bigger.

14. $\sqrt[3]{216}$

6 $6 \times 6 \times 6 = 216$

15. $m^3 = 343$

$m = 7$ $7 \times 7 \times 7 = 343$

Still one solution, for the same reason as question 8.